Introduction

Delivering computing to young learners

Oxford International Primary and Lower Secondary Computing is a complete syllabus for computing education for ages 5–14 (Years 1–9). By following the program of learning set out in this series, teachers can feel reassured that their students have access to the computing skills and understanding that they need for their future education.

Find out more at:
www.oxfordsecondary.com/computing.

Structure of the book

This book is divided into six chapters, for Year 8 (ages 12–13).

1. **The nature of technology:** Introduction to computer networks and internet services
2. **Digital literacy:** Using online research for learning and discovery
3. **Computational thinking:** Writing a program using a data structure
4. **Programming:** Using procedures or functions and comparing algorithms
5. **Multimedia:** Using technology creatively to make a video
6. **Numbers and data:** Using technology to analyse data

What you will find in each unit

- Introduction: An unplugged activity and a class discussion help students to start thinking about the topic.
- Lessons: Six lessons guide students through activity-based learning.
- Check what you know: A test and activities allow you to measure students' progress.

What you will find in the lessons

Although each lesson is unique, they have common features: learning outcomes for each lesson are set out at the start; learning content delivers skills and develops understanding.

Activity Every lesson involves a one or more learning activities for the students.

Extra challenge Activities to extend students who are able to do more.

Test A short test of four questions, of progressive difficulty, to check students' understanding of the lesson.

Additional features

You will also find these features throughout the book:

Word cloud The word cloud builds vocabulary by identifying key terms from the unit.

Be creative Suggestions for creative and artistic work.

Explore more Extra tasks that can be taken outside the classroom and into the home.

Digital citizen of the future Advice on using computers responsibly in life.

Glossary Key terms are identified in the text and defined in the glossary at the end.

Assessing student achievement

The final pages in each unit give an opportunity to assess student achievement.

- Developing: This acknowledges the achievement of students who find the content challenging but have made progress.
- Secure: Students have reached the level set out in the programme for their age group. Most should reach this level.
- Extended: This recognises the achievement of students who have developed above-average skills and understanding.

Questions and activities are colour-coded according to achievement level. Self-evaluation advice helps students to check their own progress.

Software to use

We recommend Python for writing programs at this age. For other lessons, teachers can use any suitable software, for example: Microsoft Office; Google Drive software; LibreOffice; any web browser.

Source files

You will see this symbol on some of the pages.

This means that there are extra files you can access to help with the learning activities. For example, half-completed Python programs or spreadsheet files.

To access the files, go to **www.oxfordowl.co.uk** and navigate to the 'Oxford International Secondary Programme' page then 'Oxford International Secondary Computing'.

Teacher's Guides

For more on these topics, look at the Teacher's Guide that accompanies this book.